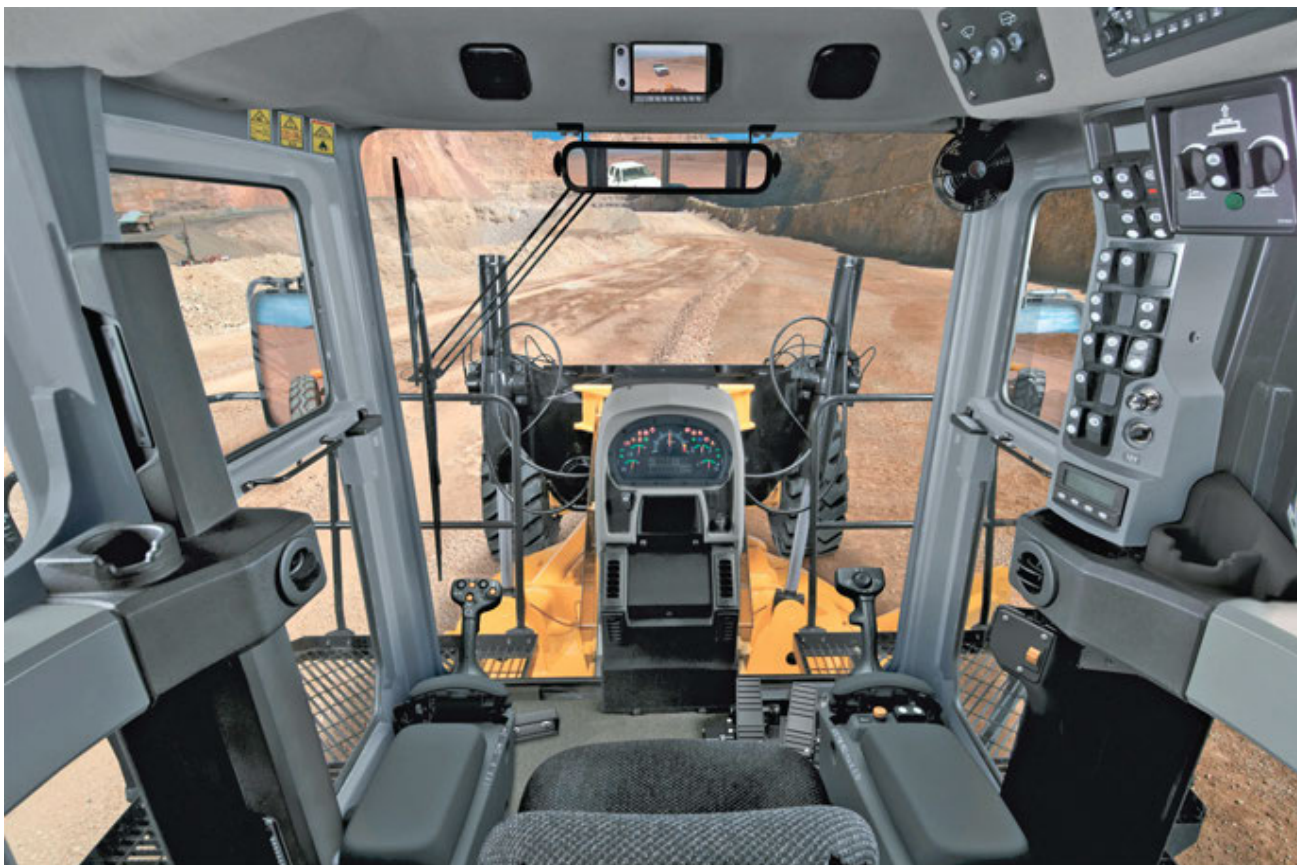




NCCER Module 22104



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Basic Operational Techniques





Section One

Prestart Procedures

Objective

1. Describe the prestart procedures for heavy equipment.
 - a. Describe how to safely mount a vehicle.
 - b. Describe common operator prestart inspection procedures for equipment, attachments, and tires.
 - c. Describe common operator prestart inspection procedures for power trains.
 - d. Describe common operator prestart inspection procedures for hydraulic systems.
 - e. Describe common operator prestart inspection procedures for electrical systems.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

There are no Performance Tasks associated with this section.



1.0.0 Prestart Procedures

- Several of the prestart activities performed on heavy equipment machinery are the same on all machines.
- Most companies have inspection checklists that specify which things need to be checked on a given machine.
- This type of inspection needs to be performed on all equipment.

Shift Inspection List			
Date _____ Submitted by _____		Repairs Needed _____	
Unit No./S.N. _____		_____	
Make/Model _____		_____	
Hour/Odometer Reading _____		_____	
Today's Use: Hours Machine... _____		_____	
• Operated _____		Additional Comments _____	
• Down (due to failure) _____		_____	
• Idle (machine not needed) _____		_____	
Use the daily inspection sheet as a guide to fill in the following information (if applicable). Explain any defects on the back side.			
	OK F R	Amount Added or *Needs Attention	
Engine Coolant	___	___	_____
Engine Oil	___	___	_____
Transmission Oil	___	___	_____
Pump Drive Oil	___	___	_____
Pivot Shaft Oil	___	___	_____
Hydraulic Oil	___	___	_____
Fuel	___	___	_____
Coolant Temp.	___	___	_____
Recorder Label	___	___	_____
Engine Air	___	___	_____
Restriction Indicator	___	___	_____
Ground Engaging	___	___	_____
Tools	___	___	_____
Tires/Undercarriage	___	___	_____
Fire Protection	___	___	_____
Seat Belt	___	___	_____
EMS/Gauges	___	___	_____
Governor Control	___	___	_____
Brakes	___	___	_____
Transmission	___	___	_____
Steering	___	___	_____
Implements	___	___	_____
Lights	___	___	_____
Horn	___	___	_____
Backup Alarm	___	___	_____
Wipers/Washers	___	___	Signature _____
Inform supervisor of any malfunction before operating machine			

1.1.0 Mounting a Vehicle Safely (1 of 2)

Operators should check the following before mounting the equipment:

- Check for any warnings or notes in the job sheet.
- Look for any markers that show where to dig or fill.
- Walk around the work area, checking the ground for large holes, large rocks, soft or muddy areas, sudden dropoffs, and curbs.
- Observe the terrain, and plan how to move the equipment safely. Move or mark sharp objects.
- Look above the site for overhead power lines, trees, and other overhangs that could contact a raised **bucket**.



1.1.0 Mounting a Vehicle Safely (2 of 2)

Observe the following safety rules when mounting heavy equipment:

- Do not use controls as handholds.
- Do not mount the machine while carrying tools or supplies.
- Do not mount a moving machine.
- Do not jump off a machine or allow others to do so.

OSHA and MSHA require that any approved seatbelt and **rollover protective structure (ROPS)** be installed on heavy equipment.



1.2.0 Inspecting Equipment, Attachments, and Tires

- Heavy construction equipment is normally mounted on an undercarriage riding on either metal tracks or rubber tires.
- Structural hardware includes the undercarriage and all things mounted to it, as well as the machine's body.



1.2.1 Tires

- Check tires for bulges, cuts, punctures, and signs of excessive wear.
- Check the wheels to ensure that lug nuts are in place and properly torqued.
- Verify that the inflatable tires are properly inflated.
- Tire inflation pressures are often posted near each tire.



INFLATABLE TIRE



SOLID RUBBER TIRES

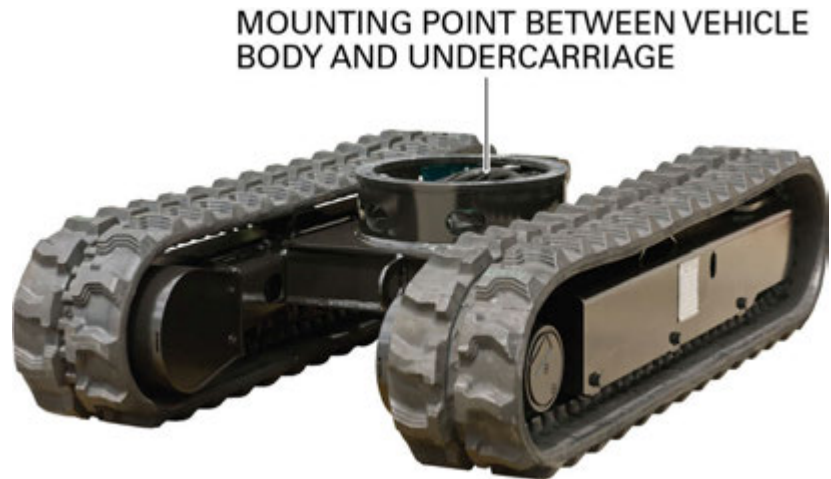
1.2.2 Tracks (1 of 2)

- Preoperational checks need to include making sure that all debris is removed from the tracks.
- Look for loose, damaged, or missing bolts or pins that hold the track shoes to the chain.
- Check proper track tension per the manufacturer's instructions.
- Look for loose or damaged track shoes.



1.2.2 Tracks (2 of 2)

- Some lighter-weight construction vehicles have rubber tracks.
- The rubber tracks are one large rubber belt with molded cleats.
- Make sure that the track appears to be properly installed and tensioned per the manufacturer's instructions. The track should not be excessively worn, ripped, or torn.



1.2.3 Rotating Parts

- Bearings must be either greased or bathed in some kind of lubricant (oil).
- If bearings need to be greased, make sure the grease fittings are cleaned before attempting to add grease to them.
- Make sure that the right kind of lubricant is being used.



1.2.4 Frame and Body Panels

- Keeping the main protective structure around and over the operator in good condition is critical to the safe operations of the machine.
- Check for cracks in the metal (especially the frames), loose connections (bolts or other types of fasteners), broken or cracked welds, and excessive corrosion.
- Dirt, mud, and moisture collect on the surfaces of the machine's body panels, ladders, and walking surfaces, and must be removed.
- Make sure ladders and walking surfaces are clean.



1.2.5 Safety Equipment

- Check all installed or mounted safety equipment on the vehicle including mirrors, seat belts, horns, and the fire extinguisher.
- Verify that lights are properly mounted and not damaged.
- Check that all brush guards are properly mounted and undamaged.
- Make sure windows are clean and undamaged.



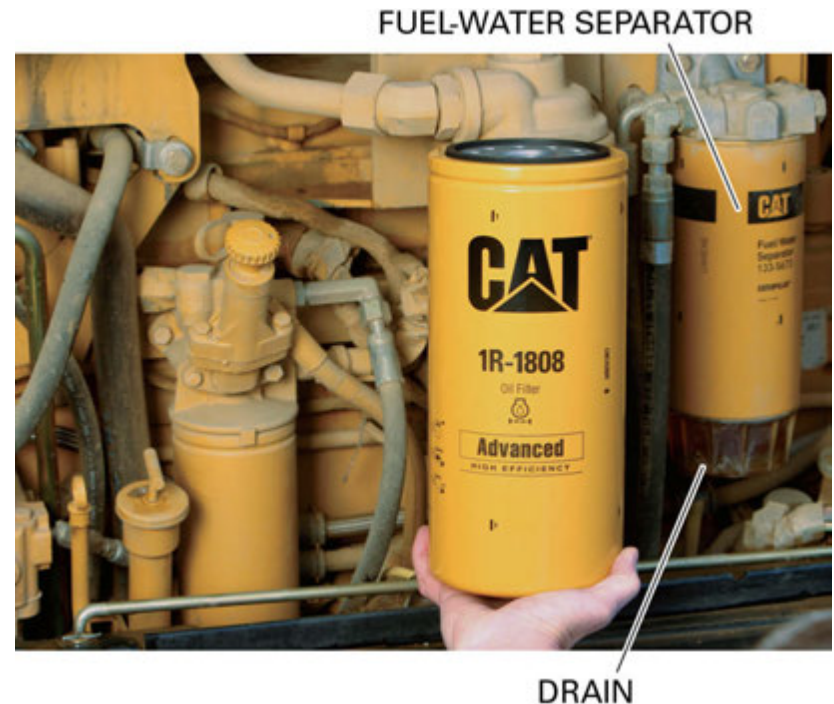
1.3.0 Power Train Inspections and Checks

Regardless of engine type, the operator needs to visually check the machine's fuel supply system, coolant system, engine, transmission, drive axle(s), and hydraulic system for signs of leaks.



1.3.1 Fuel Supply

- Fuel filters must be changed.
- Fuel filters on heavy equipment machines are usually mounted near the engine's fuel injector (or carburetor).
- Check for visible water in the fuel-water separator and drain it into an approved container.
- Never fill the fuel tank or work on any part of the fuel system with the engine running, while smoking, or near an open flame.



1.3.2 Coolant System Check



SWING OUT RADIATOR COOLING FAN



RADIATOR CAP INSIDE PROTECTIVE COVER

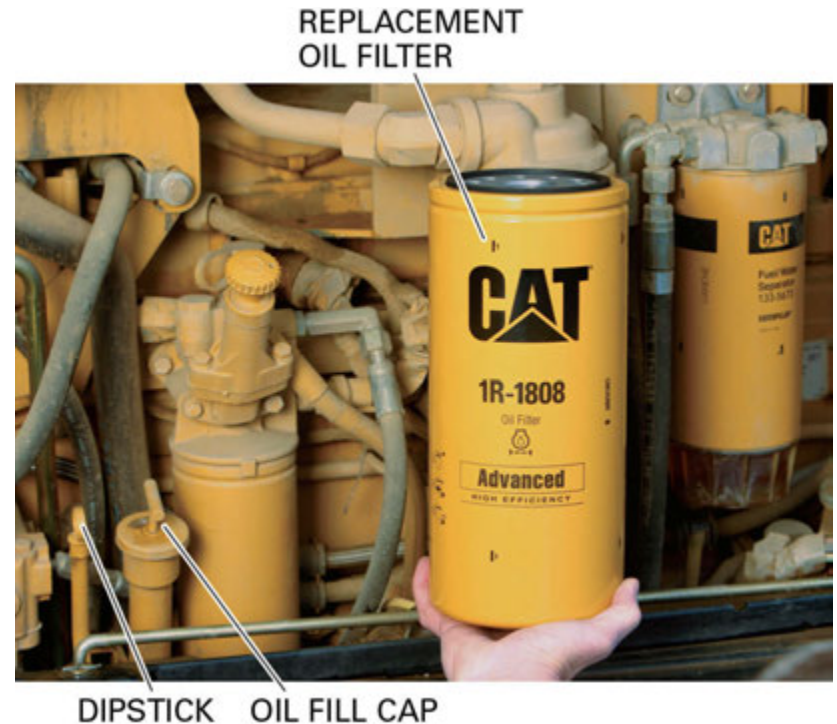


COOLANT TANK INSIDE MACHINE

- Start the power train inspections at the engine's radiator.
- Make sure that the air paths into, through, and out of the radiator are free of debris.
- The hoses between the engine and the radiator must be securely fastened and in good condition.
- If the engine is cool, remove the radiator cap and check the fluid level in the radiator.

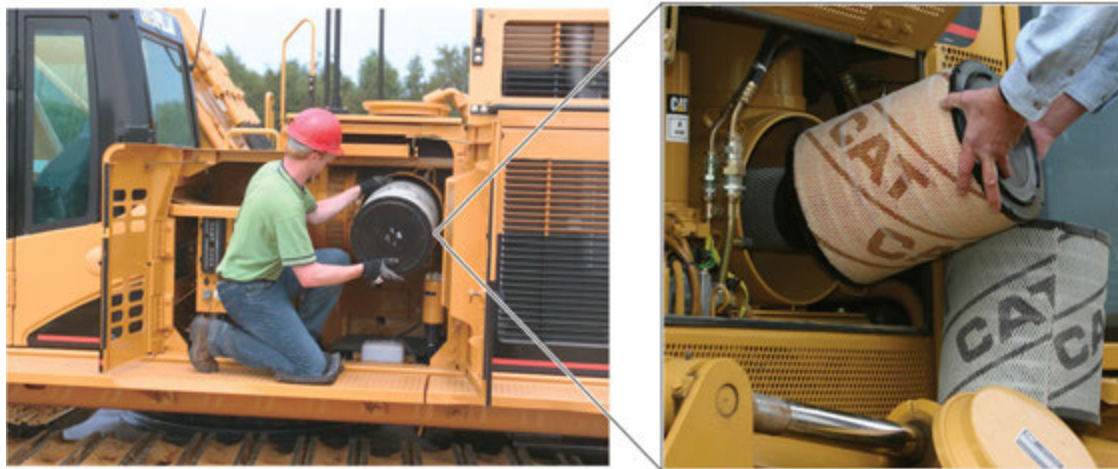
1.3.3 Engine Oil Check

- Dipsticks have markers that show where the oil level should be, and a level where oil needs to be added.
- Make sure that only the oil called for in the machine's O&M manual (or company procedures) is used.
- Do not overfill the oil.
- Change the oil filter according to the manufacturer's maintenance schedule.



1.3.4 Engine Air Filter Check

- Most heavy equipment engines have large air filters housed inside of metal canisters mounted somewhere near the engine.
- In most cases, the filter can be slipped out for visual inspection. Many filters can be cleaned if they are not too oily and dirty. If in doubt, replace any dirty filter.



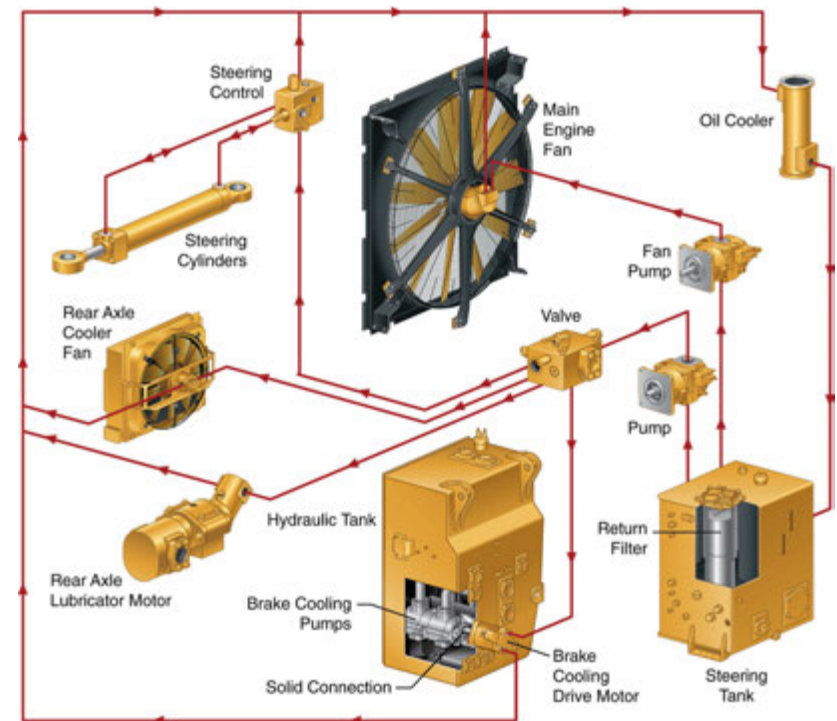
1.3.5 Transmission and Drive Axle Checks

- Transmissions and drive axles need lubricants to protect the gears.
- Any transmission, other than a standard shift transmission, also requires a specific transmission fluid in order to function as designed.
- Some transmissions have sight glasses that indicate the level of fluid in the transmission. Others may have a dip stick.
- Drive axles usually have inspection ports that can be opened to check the lubricant level.
- Some modern machines may have temperature and level sensors in or on both the transmission and drive axle(s).



1.4.0 Hydraulic System Inspections and Checks

- Almost all heavy equipment vehicles use a hydraulic system to control their attachments.
- A hydraulic system is a closed-loop system consisting of the reservoir, a pump, filters and strainers, control valves, hoses, and actuators (cylinders or motors).
- Hydraulic power may also drive cooling fans and lubrication pump motors.



1.4.1 Hydraulic System Reservoir and Pump (1 of 2)

- Most hydraulic reservoirs have a sight glass that allows the level of fluid to be checked without opening the reservoir.
- If hydraulic fluid needs to be added, it can be poured in through the fill port located on the reservoir.
- Most fill ports have a strainer or filter inside them to keep the fresh hydraulic fluid clean.



HYDRAULIC
FLUID FILL PORT

HYDRAULIC FLUID
SIGHT GLASS

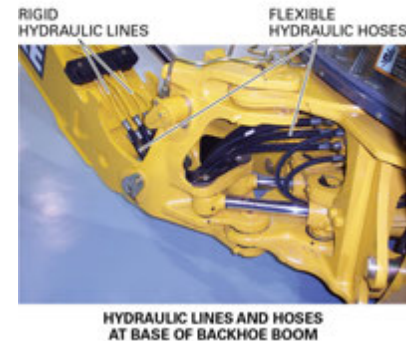
1.4.1 Hydraulic System Reservoir and Pump (2 of 2)

- The temperature of hydraulic fluid rises as it is forced through the hoses and control valves to make the actuators perform work.
- Hydraulic fluid may pass through a filter before returning to the reservoir.
- A heat exchanger may provide additional cooling for the returning hydraulic fluid.



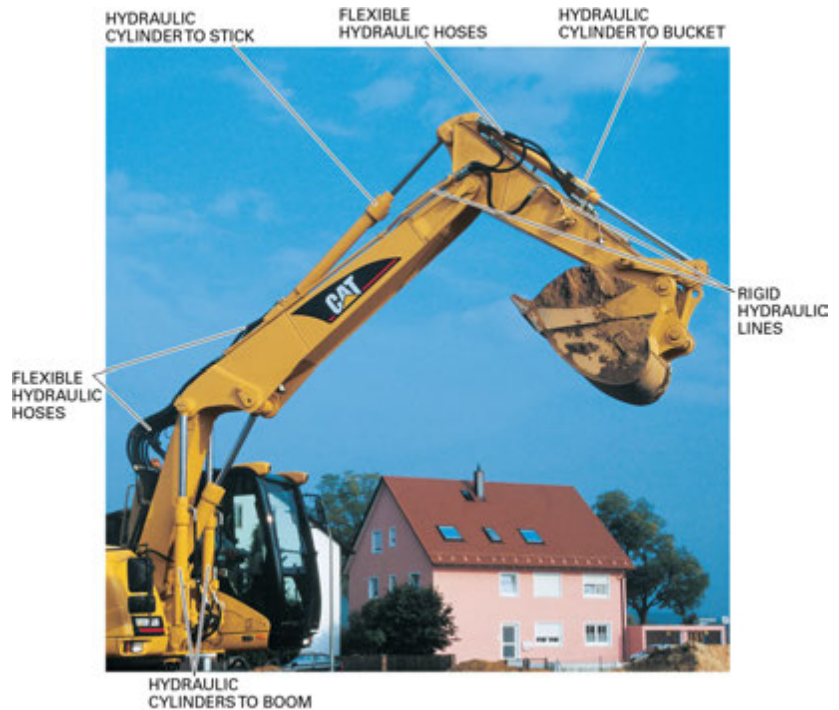
1.4.2 Hydraulic System Hoses (1 of 2)

- Check the metal connectors that connect the hydraulic hoses to the body of the pump.
- Make sure that the hose connectors are securely fastened to the pump's body. Always check the hoses for cracks and wear spots.
- The points where the flexible hoses and the rigid lines connect should always be checked for leaks.



1.4.2 Hydraulic System Hoses (2 of 2)

- The hydraulic lines and hoses mounted inside the machine's protective housing are less likely to be exposed to potential damage than those installed out on the boom and stick of a backhoe or excavator.
- Check every hose or line that can be seen. The hoses and lines on the movable attachments are more likely to get damaged.



1.5.0 Electrical and Control Inspections and Checks

- Visually check the control panels for broken glass or gauges that appear to be broken.
- If allowed, turn the On/Off switch to On and recheck all the gauges and indicators on the control panel being inspected.



1.5.1 Batteries and Cables (1 of 2)

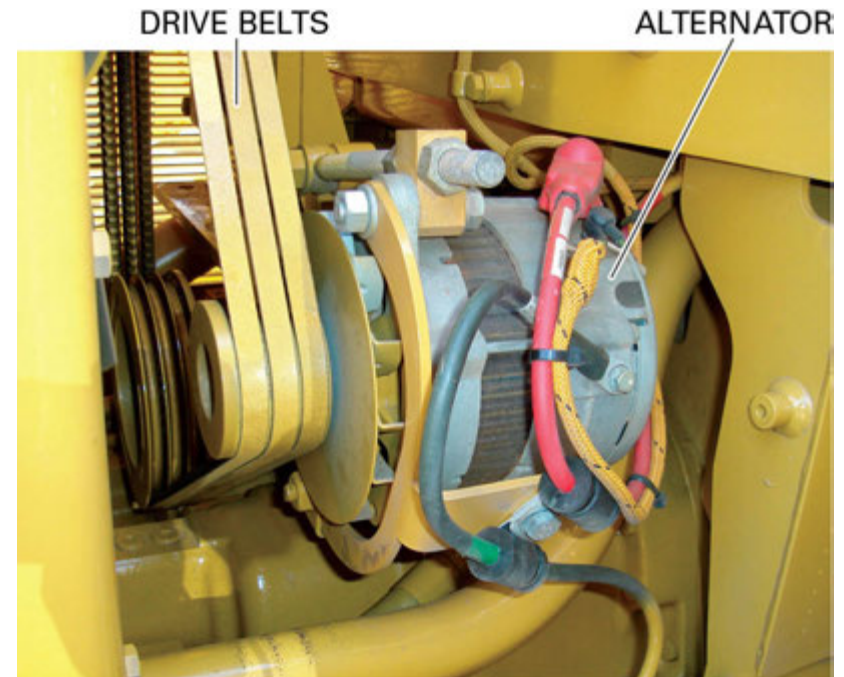
Heavy equipment engines require more battery power to start, so the heavy equipment vehicles are equipped with multiple batteries.



RED POSITIVE TERMINALS

1.5.1 Batteries and Cables (2 of 2)

- Make sure that all cables are secured and undamaged.
- Look for corrosion on the terminals.
- Verify that the alternator and all its terminals appear to be securely mounted and undamaged.
- When the engine is started, a current or amperage meter should indicate that the alternator is recharging the batteries.



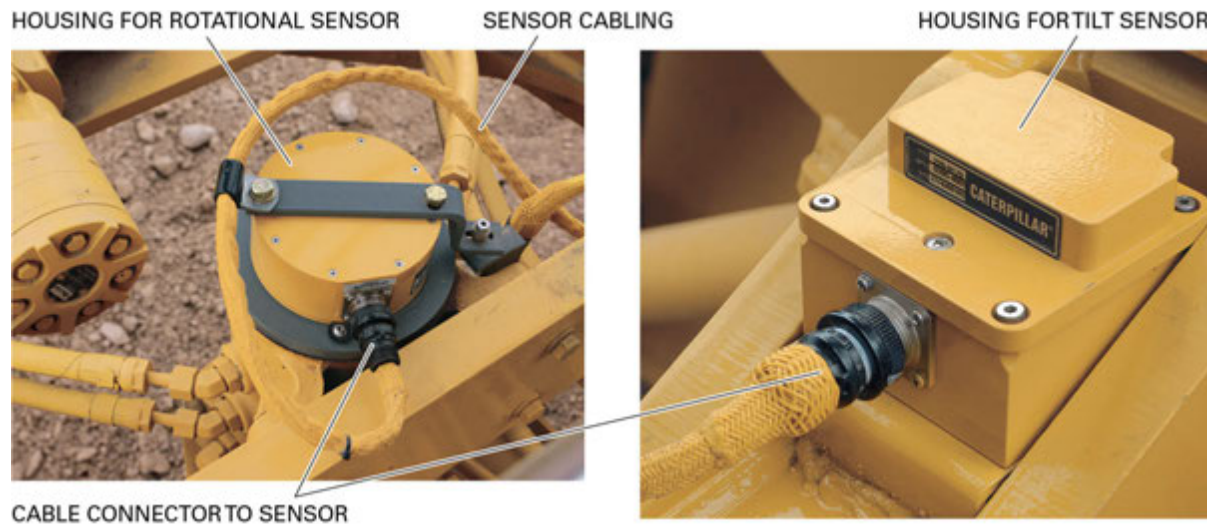
1.5.2 Lights, Horn, and Alarms

- Visually inspect any installed lights to make sure that they are undamaged, clean, and securely fastened.
- Make sure that lights appear to be pointing in the proper direction.
- Verify that the wiring into the lights is securely fastened and undamaged.
- Check the machine's horn and any alarms.



1.5.3 Sensors

- On construction equipment, sensors tell the computer how the devices attached to the machine are positioned (raised, lowered, or at some angle) as well as basic vehicle information.
- Verify that sensors and cabling are securely attached and undamaged.



Wrap Up – Trade Terms

Bucket: A U-shaped, closed-end scoop attached to a front-end loader or backhoe.

Rollover protective system (ROPS): A metal structure attached to the equipment's cab designed to protect the operator should the equipment roll over.



1.0.0 Section Review Question 1

1. The minimum safety gear that should be provided on a heavy equipment machine are seat belt, fire extinguisher, and _____.
- a. a hard hat
 - b. a ROPS
 - c. a fall protection harness
 - d. safety goggles



1.0.0 Section Review Question 1 Answer

1. The minimum safety gear that should be provided on a heavy equipment machine are seat belt, fire extinguisher, and _____.
- a. a hard hat
 - b. a ROPS**
 - c. a fall protection harness
 - d. safety goggles



1.0.0 Section Review Question 2

2. Metal tracks are made up of track shoes fastened to a _____.
- a. sprocket
 - b. hydraulic actuator
 - c. track chain
 - d. roller



1.0.0 Section Review Question 2 Answer

2. Metal tracks are made up of track shoes fastened to a _____.
- a. sprocket
 - b. hydraulic actuator
 - c. track chain**
 - d. roller



1.0.0 Section Review Question 3

3. In *Figure SR01*, the operator is holding a _____.
- a. fuel filter
 - b. fuel-water separator
 - c. oil filter
 - d. hydraulic reservoir



Figure Credit: Reprinted courtesy of Caterpillar Inc.

1.0.0 Section Review Question 3 Answer

3. In *Figure SR01*, the operator is holding a _____.
- a. **fuel filter**
 - b. fuel-water separator
 - c. oil filter
 - d. hydraulic reservoir



Figure Credit: Reprinted courtesy of Caterpillar Inc.

1.0.0 Section Review Question 4

4. The arrow in *Figure SR02* is pointing at the _____.
- a. engine oil fill port
 - b. fuel fill port
 - c. Coolant fill port
 - d. hydraulic fluid fill port



Figure Credit: Reprinted courtesy of Caterpillar Inc.

1.0.0 Section Review Question 4 Answer

4. The arrow in *Figure SR02* is pointing at the _____.

- a. engine oil fill port
- b. fuel fill port
- c. Coolant fill port
- d. hydraulic fluid fill port**



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1.0.0 Section Review Question 5

5. The batteries on a heavy equipment machine are regularly charged by ____.
- a. jump starting
 - b. a portable battery charger
 - c. the engine alternator
 - d. the hydraulic system



1.0.0 Section Review Question 5 Answer

5. The batteries on a heavy equipment machine are regularly charged by _____.
- a. jump starting
 - b. a portable battery charger
 - c. the engine alternator**
 - d. the hydraulic system



Next...

**Section 2.0.0
Starting, Operating, and Shutting
Down Heavy Equipment**



*Read Sections 2.0.0 through 2.3.0 and
complete the Section Review.*